

$$\begin{array}{ccc}
 \widehat{\mathbb{C}} \setminus f^{-1}(C) & \xrightarrow{f} & \widehat{\mathbb{C}} \setminus C \\
 \cup & & \cup \\
 U & \longrightarrow & V
 \end{array}
 \left. \vphantom{\begin{array}{ccc} \widehat{\mathbb{C}} \setminus f^{-1}(C) & \xrightarrow{f} & \widehat{\mathbb{C}} \setminus C \\ \cup & & \cup \\ U & \longrightarrow & V \end{array}} \right\} \text{component-wise}$$